B. K. Birla College of Arts, Science and Commerce, Kalyan (W)

TYBSc Physics, Semester VI Special Theory of Relativity

Assignment 3

Unit 2: Magnetism as a Relativistic Effect and the Transformation of Fields

Name: _____ Roll No.: _____ Date: _____

Total marks: 60 **Each question carries 5 marks**

Instructions: Attempt all questions. Section A: theory (write to the point, with neat diagrams where useful). Section B: numerical problems and derivations (state formulae used, show all steps and give units). Sign convention for the interval: $s^2 = \Delta x^2 + \Delta y^2 + \Delta z^2 - c^2\Delta t^2$; four-vectors $a^\mu = (a^0, \mathbf{a})$ with $a \cdot b = -a^0 b^0 + \mathbf{a} \cdot \mathbf{b}$.

Section A: Theory (6 × 5 = 30 marks)

Q1. "Classical electrodynamics is already consistent with special relativity." Explain this statement using the example of a magnet and a conducting loop in relative motion. **[5]**

Q2. Explain the interdependence of electric and magnetic fields. Why are $\mathbf{E}$ and $\mathbf{B}$ not separate physical entities? What is meant by the statement that electric charge is invariant? **[5]**

Q3. Explain qualitatively how a current-carrying wire, which is neutral in the laboratory, appears charged in the rest frame of a charge moving parallel to it, and how this accounts for the magnetic force on the charge. **[5]**

Q4. The drift speed of electrons in a wire is only about $10^{-4}\,\mathrm{m/s}$, so $v^2/c^2 \sim 10^{-25}$. Explain why the magnetic force between two current-carrying wires is nevertheless easily observed. **[5]**

Q5. Using a parallel-plate capacitor and a long solenoid, explain why the components of $\mathbf{E}$ perpendicular to the direction of motion change under a Lorentz transformation while the parallel components of $\mathbf{E}$ and $\mathbf{B}$ do not. Write the complete set of field-transformation equations. **[5]**

Q6. Explain the two field invariants $\mathbf{E} \cdot \mathbf{B}$ and $E^2 - c^2B^2$. Using them, discuss when an electric (or magnetic) field can be made to vanish by a suitable choice of frame. **[5]**

Section B: Numericals and Derivations (6 × 5 = 30 marks)

Q7. A neutral wire is modelled as a line charge $+\lambda$ moving with speed v and a line charge $-\lambda$ moving with speed v in the opposite direction. A charge q moves parallel to the wire with speed u at a distance s. Using the net line charge in the rest frame of q, $\lambda_{\mathrm{tot}} = -2\lambda uv/\big(c^2\sqrt{1-u^2/c^2}\big)$, derive the force on q in the laboratory and show that it equals quB with $B = \mu_0 I/2\pi s$. **[5]**

Q8. Using a parallel-plate capacitor, derive the transformation rules $E^\perp = \gamma E_0^\perp$ and $E^\parallel = E_0^\parallel$ for a field produced by charges at rest in S_0. **[5]**

Q9. In frame S there is a uniform electric field $\mathbf{E} = 5.0 \times 10^5\,\hat{\mathbf{y}}$ V/m and no magnetic field. Find $\bar{\mathbf{E}}$ and $\bar{\mathbf{B}}$ in a frame moving at $0.8c$ along x, and verify that $E^2 - c^2B^2$ is unchanged. **[5]**

Q10. Using the field-transformation equations for a boost along x, prove that $\mathbf{E} \cdot \mathbf{B}$ is a Lorentz invariant. **[5]**

Q11. In frame S, $\mathbf{E} = 4.0 \times 10^4\,\hat{\mathbf{y}}$ V/m and $\mathbf{B} = 2.0 \times 10^{-4}\,\hat{\mathbf{z}}$ T. (a) Show that there is a frame moving along x in which the electric field vanishes, and find its speed. (b) Find the magnetic field in that frame. (c) Check your answer with the invariant $E^2 - c^2B^2$. **[5]**

Q12. A long straight wire carries a current of 5.0 A. A proton moves parallel to the wire, in the direction of the current, at 1.0×10^5 m/s and a distance of 2.0 mm from it. (a) Find the magnetic force on the proton in the laboratory. (b) In the proton's rest frame the wire carries a net line charge $\lambda' \approx -uI/c^2$. Calculate λ', the electric field it produces at the proton, and the electric force, and compare with (a). **[5]**

Useful constants: $c = 3.0 \times 10^8$ m/s; $e = 1.6 \times 10^{-19}$ C; $m_e c^2 = 0.511$ MeV; $m_p c^2 = 938.3$ MeV; $\varepsilon_0 = 8.85 \times 10^{-12}$ F/m; $\mu_0 = 4\pi \times 10^{-7}$ H/m; $1/(4\pi\varepsilon_0) = 9.0 \times 10^9$ N m^2/C^2; $G = 6.67 \times 10^{-11}$ N m^2/kg^2; $M_\odot = 1.99 \times 10^{30}$ kg; 1 ly $= 9.46 \times 10^{15}$ m; 1 pc $= 3.086 \times 10^{16}$ m $= 3.26$ ly; 1 yr $= 3.156 \times 10^7$ s; $b_{\text{Wien}} = 2.898 \times 10^{-3}$ m K; $k_B = 8.62 \times 10^{-5}$ eV/K; $a = 7.566 \times 10^{-16}$ J m^{-3} K^{-4}.